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1. INTRODUCTION

1.1 Purpose

The purpose of this procedure is to provide an overall set of instructions on how to use the Maintenance Ticket Tracking System (MTTS) for all Satellite Network Operations Center personnel. The Maintenance Ticket Tracking System (MTTS) is an application that is used to document subscriber and system anomalies, distribute and track Configuration and Change Order implementation, and track corrective and scheduled maintenance throughout the Iridium Communication System (ICS).

1.2 Scope

This document is for use by all Satellite Network Operations Center personnel that monitor and report the status of the Iridium Communications System (ICS). The MTTS system is used as a means of communication throughout the ICS to include reporting outages, tracking problems, requesting information, N/CCO implementation, and maintenance activities.

1.3 Related Documents

<u>Document Title</u>	<u>Document ID:</u>
FTP Operations	12-P40233L

1.4 Acronyms:

ARS – Action Request System
CCO – Configuration Change Order
DSO POC – Distributed Server Option Point of Contact
FTP – File Transfer Protocol
GTAC – Gateway Technical Assistance Center
GTO – Gateway Technical Office
GW – Gateway
ICS – Iridium Communication System
ILAP – Iridium Link Access Protocol
MO – Maintenance Order
MTTS – Maintenance Ticket Tracking System
NIP – Network Implementation & Planning
ODN – Operational Data Network
SNOC – Satellite Network Operation Center
TT – Trouble Ticket
WLD – Work Log Description

2. PROCEDURE

2.1 HOW TO ACCESS MTTS FROM THE SNOG ENVIRONMENT

*NOTE: The following steps assume you are already logged in to a UNIX workstation.

- 2.1.1 From the User window, initiate a terminal window session. You can do this by clicking your right mouse button on a blank area of your screen. You will then see a menu list, use your mouse and scroll down to Programs, then move mouse over to Terminal and left click on the word Terminal. The menu list will disappear and a Terminal window will automatically populate.
- 2.1.2 From your user directory (/home/xxxx_x), type `xhost +mcf_mtts` (NOTE: a space exist between `xhost` and `+mcf_mtts`. This will allow your workstation to display a window from the MTTS server)
- 2.1.3 NOTE: Due to different workstations being on different LAN's, accessing MTTS has some slight differences. Please see the below information to determine which access option you need to follow in order to connect to MTTS.

Mission LAN's and EA LAN workstations: use `telnet mtts`

Not LAN workstations: use `mtts_start`

ODN workstations: use `telnet mcf_mtts`

- 2.1.4 This will bring up a login and password screen (Note: This will be giving you access into the Remedy server). Based on your access into the server, this will determine the login and password you should be using (see below):

- 1. Mission/EA LAN: use - login: `mtts` password: `mtts`
- 2. Not LAN: use - login: `mtts2` password: `mtts2`
- 3. ODN workstations: use - login: `mtts` password: `mtts`

2.2 LOGIN TO THE ACTION REQUEST SYSTEM AND NOTIFICATION TOOL

- 2.2.1 Action Request System - Section 2.1 has given you access into the Remedy server, you now need access into the MTTS application. You should now see an Action Request System and Notification Tool screen, to include an associated login screen for each. In

order to log into the MTTTS applicaton and take action on a ticket, you need to know what action is to be taken – see table below:

Action to be taken:	Login needed:
Creating a Trouble/Maintenance Ticket	Your personal login provided by administrator
Acknowledging a ticket sent to the SNOC from an outside entity (GW (GTO), BSC, etc.)	DSO POC (password known by Network Lead)
Clearing ticket back to originator (Trouble ticket or Maintenance Order)	You need to be logged in as whomever is listed in the Current Assignee field. (*Note: if the person listed in the Current Assignee field is not currently on shift you can log in as netops_s, which has manager privileges, and work this ticket on their behalf. The Network Lead can provide you the password for this login).
Closing an IMCF ticket	Originator of the ticket or netops_s (netops_s has manager privileges)
Change Current Assignee for internal routing	Must be logged in as the entity listed in Current Assignee field or if assigned to an individual that is part of the Net Ops team, you will be able to work this ticket as netops_s. (netops_s has manager privileges)

2.2.2 Notification Tool – As stated above, once you have accessed the MTTTS application a login screen will appear for the Notification Tool. In order to receive notification when a Trouble Ticket (TT)/Maintenance Order (MO) has been received on the server or assigned to you, you must be logged into the Notification Tool. You must use the login and password of the entity from which you wish to be notified.

2.2.3 NOTE: The main log in for MTTTS for tickets entering the SNOC server is DSO POC. The first person to log in as DSO POC will receive these notifications. Anyone else that attempts to login as DSO POC will be able to do so, but will not receive any notifications. This login is restricted for the Network Lead position or an individual designated by the Network Lead to monitor and acknowledge this queue. The MTTTS system is designed to alert you of the assignment of a TT/MO the instant the Current Assignee field has been changed and the Apply button is depressed. (Example: A ticket is transferred from INA-H to the SNOC, via the DSO POC option, in order to receive this notification you must be logged in as DSO POC).

2.3 SETTING UP USER MTTTS PREFERENCES

The MTTTS Preferences are configurable to meet the needs and desires of each individual user. Each user can set options for controlling the appearance and behavior of various aspects of the User Tool. Once you have set the desired preferences and click the 'Apply' button, these changes are written to your personal login directory and stored for you the next time you log in. See steps below:

2.3.1 Here are some of the Preferences you can choose from and what they mean:

- ☐ Max in Active Schema List – Specifies the maximum number of schemas kept in the active schema list is displayed at the bottom of the File menu. (Default value is 4).
- ☐ Max in Query – Specifies the max number of tickets that are returned to you when performing a query menu operation. (The default value is 1000, to set no limit for the number of entries enter 0).
- ☐ Prior to New Query – **Clear** – Clears all information from the fields on the schema. (This is the default). **Set to Defaults** - Loads the schema fields with any existing default information. **Keep previous** – Retains information entered from your current or last query.
- ☐ Query Bar – **Show** – makes the query bar visible (This is the default). **Hide** – Hides the query bar.
- ☐ Double-click action on Query – **Display** – causes a Display window to open after you double-click a query list item. (This is the default). **Modify** – causes a Modify Individual window to open after you double-click a query list item. **Report** – causes a Report window to open after you double-click a query list item.
- ☐ Prior to New Submission – **Clear** – clears all information from the fields on the schema. (This is the default). **Set to Defaults** – loads the schema fields with any existing default information. **Keep Previous** – retains information from the last submitted AR.
- ☐ Confirmation Upon Successful Submit – **Footer Only** – displays only a footer message after a successful AR submission. (This is the default). **Confirmation Box and Footer** – displays both a footer message and a confirmation dialog box after a successful AR submission.
- ☐ Submit Window-Text/Edit Diary window Apply action – **Leave Window Up** – leaves the window displayed after you click Apply on a Text Edit or Diary Entry dialog box opened from a submit window. **Close Window** – closes the window after you click Apply on a Text Edit or Diary Entry dialog box opened from a Submit window. (This is the default).

2.3.2 Other Windows-Text/Edit Diary window Apply action – **Leave Window Up** – leaves the window displayed after you click Apply on a Text Edit or Diary Entry dialog box opened from a query, Modify Individual, or Modify All selected window. (This is the default). **Close Window** – closes the window after you click Apply on a Text Edit or Diary Entry dialog box opened from a query, Modify Individual, or Modify All selected window.

2.4 SETTING UP MTTTS NOTIFICATION TOOL PREFERENCES

Users can modify their MTTTS Notify Tool preferences. They can instruct MTTTS how they want to be notified when a Trouble Ticket (TT)/Maintenance Order(MO) has been assigned/received. You can specify whether to save notifications between runs, the maximum number of notifications in the notification window, where to log the notifications, and whether they want the display or modify command type. The operator must first log into the MTTTS Notifier Tool as stated above in Section 2.2. The operator can then choose the options they would like set from the Properties window. See. steps below:

- 2.4.1 From the Notifier tool line menu select **File → Options → Properties**. The Properties window is displayed.
- 2.4.2 The User may select several combinations from the Notification Tool Properties window:
 - ☐ Action on New Message – specifies how the Notification Tool reacts when a notification arrives: (Flash, Beep, Open, Open Tool)
 - ☐ Save Notifications Between Runs – select to save notifications between sessions of the Notification Tool.
 - ☐ Maximum Notifications in Window – specifies the maximum number of notifications that you can save and display in the Notification Tool window.
 - ☐ Log Notifications to File – specifies a path and file name to be used as the audit trail log file for all notifications. If you do not specify a name, no audit trail file is kept.
 - ☐ Command Type – specifies whether a Display or a Modify Individual window appears when you select Get Details for or double-click a notification that is related to an AR (Display, Modify)
- 2.4.3 The Notification Tool Properties window will reflect changes made by the operator.
- 2.4.4 Select "Apply", preferences are saved, return to Notification Tool, or Select "Dismiss", preferences are discarded, return to Notification Tool.
- 2.4.5 NOTE: The Properties setup is configurable by each user. It is suggested that the user at least check the blocks for Flash, Beep, and Open under Section 'Action on New Message' to ensure timely notification.

2.5 QUERYING THE SNOC DATABASE FOR A MTTTS TICKET

This section will take you through the steps for searching the SNOC database for a specific ticket(s) in order for the user to take action on a specific ticket(s). See steps below:

- 2.5.1 NOTE: This section assumes you have completed Section 2.2 and have access into the MTTTS application.
- 2.5.2 Querying the SNOC Database for a MTTTS ticket can be done in one of two ways: First, by querying in syntax specific text from the Query bar at the bottom of the main ARS screen ('MT ID:' LIKE "IMCF00000XXXX"). When using the Query bar, field names must be listed between single quotations and data you are querying from that field name must be in double quotations. Secondly, by inputting exact data into the fields from which you want information returned (MTTS ticket number would be inputted into the MT ID: field).
- 2.5.3 NOTE: In order to get data to populate into the Query bar, you can do this one of two ways. First, click on the actual lettering of that specific field (the field name will automatically show up in the Query bar) in the MTTTS schema. Secondly, click on the FIELDS list menu button, above the Query bar, and choose the specific field name from the field names that are listed. (The key to this operation is to give the name of the field you want to query, then give the data within that field you want the system to find for you).
- 2.5.4 NOTE: Queries are resource intensive and cause higher utilization of the CPU on the server. When doing a query to retrieve the data that you need, make the query as specific as possible. When inputting data into a specific field (Example: a persons name in the Originator field), your output will be every single ticket on the server that has the information that you have specified for this field.
- 2.5.5 Once you have entered the information needed. Go to the ARS tool line menu, select **Query → List** (for a quick brief description of the ticket in a List mode) or Modify Individual, to open the ticket up in a Modify Individual window.
- 2.5.6 Once you have the ticket open in a Modify Individual window, you may now update and take action on the MTTTS ticket, as needed. Having a ticket open in the Modify mode, is the key to updating any ticket within the SNOC server.

2.6 OPEN (CREATE) A TROUBLE TICKET (TT)

A situation has arisen that requires a Trouble Ticket to be opened. Examples are: a detected and validated network affecting fault (i.e.: Abnormal ILAP termination, ODN circuit being down, etc.), the need to perform preventative maintenance, a request to another ICS entity to perform a needed action, or a request for information. (Please see Appendix A for list of codes to be used when opening/creating a Trouble ticket). See steps below:

- 2.6.1 NOTE: This section assumes you have access into the MTTTS application as described in Section 2.2. Once you have gained access into the MTTTS application, you will now be

prompted for a schema to work from. In most cases, the only schema to be used is: MTTTS:MaintenanceTicket (mcf_mmts) schema.

- 2.6.2 From the Action Request System window tool line menu, select **File → Open Submit** menu item. A new window will be populated in the Submit mode. (Notice: At the top of the new window, Submit will be displayed).
- 2.6.3 Enter data into the fields of the Submit window of the Trouble Ticket (TT).
- 2.6.4 NOTE: Fields with bold face typeset are required fields. The required fields are: *Originator, System/Service, Subsystem/Feature, Trouble/Maintenance Type, Work Log Description(English), Initial Affected Element ID, Priority, MT State, MT Status, Work Performed, and Current Assignee*. Users must populate the correct data, to reflect the reason for creating this ticket, into all required fields in order for the TT to be successfully submitted. (If for some reason the data you need to use to create the Trouble Ticket is not present, please inform the MTTTS System Administrator via e-mail with specific facts as soon as possible. Otherwise, use the code that is the most closely related).
- 2.6.5 NOTE: If the Trouble Ticket being created is for another entity, outside the SNOC environment, this will be a Transfer. If the Trouble Ticket being created is to be handled by another user/engineer within the SNOC environment, this is not a Transfer, but Assigning the Ticket (see *Sections 2.9 Assigning a Trouble Ticket(TT)/Maintenance Order(MO) & 2.10 Transfer a Trouble Ticket(TT)/Maintenance Order(MO)*, along with Appendix B to assist in understanding the differences).
- 2.6.6 Click on the 'Apply' button. The data is saved and the MTTTS application will return to you a message stating the MT ID number that has been created. If you transferred the ticket to another server, the MTTTS application will inform you of the ticket being transferred and the server that that the system transferred the ticket to. This message should match the server you intended the ticket to be transferred to. If it does not, one of two things may be the problem. One, an operator error has occurred and the ticket was transferred to the wrong server by accident. Two, possibly the mapping for this gateway has been corrupted. If you are unsure of the problem contact the MTTTS administrator to assist you.

2.7 OPEN (CREATE) A MAINTENANCE ORDER(MO) - CHILD TICKET

A Maintenance Order (MO-Parent Ticket) has been received at the SNOC from NIP or GTAC informing of an update/change that needs to be accomplished within a certain GW(GTO). These Parent (MO) tickets are delivered to the SNOC with the contents of either a NCCO (NIP Configuration Change Order - from NIP) or a CCO (Configuration Change Order – from GTAC). The text of these documents will be pasted in the Work Log Description of each individual ticket then transferred to the SNOC. The GW that this N/CCO (MO) is destined to be

delivered to will be listed within the text pasted in the Work Log Description within the Parent Ticket-MO. For all N/CCO's that will cause a service affecting outage to a GW component and will be customer impacting, must have a SIR(Service Impact Request) submitted and approved prior to ftp'ing any files to the GW. The process of accomplishing this will be explained in the steps below. See steps below:

- 2.7.1 Incoming MO to the SNOC server - The Parent Ticket-MO will be sent to the SNOC from an outside entity (GTAC or NIP). In order for you to receive Notification of this new ticket now on the SNOC server, you must be logged into the Notification Tool as DSO POC. (Please see *Section 2.4 – Setting up MTTS Notification Tool Preferences* to understand the responsibilities of the DSO POC role).
- 2.7.2 Acknowledge the MO - The Network Lead or designated representative monitoring the DSO POC queue must add a note in the Work Log Description of acknowledging this Parent Ticket-MO. The statement that needs to be in the Work Log Description must be in one of these two formats: "Acknowledging ticket," or "ACKNOWLEDGING TICKET." Along with this note, you can include your next plan of action (i.e.: Acknowledging ticket, printing ticket for Iridium approval). *NOTE: It is mandatory that every user that updates the Work Log Description, sign their name after the notes they have included to inform who is putting in the note and what action that person has taken.
- 2.7.3 Verify the Parent Ticket-MO is in the correct format - This N/CCO must be in a certain format and have the proper approval signatures included, in order for this MO to be processed and a Child ticket created. You should consult with the NL (Network Lead) or Iridium Lead for additional training on the correct format the N/CCO needs to be received in for proper processing to the GW. This prior approval must be done prior to sending any files or tickets to the specified GW.
- 2.7.4 Inform Iridium of MO received – Once you have acknowledged the Parent Ticket-MO, you must then inform the Iridium Lead or designated representative of the Parent Ticket-MO being received. You may do so by printing a copy of the Parent Ticket-MO and present it to the Iridium operator. The Iridium personnel will add this MO to their list and check with their management for approval to create a Child Ticket-MO and send it to the appropriate GW.
- 2.7.5 SIR Process - For all N/CCO's that will cause a service affecting outage, the N/CCO will have to go through the SIR (Service Impact Request) process. This process is to be followed prior to sending any files or ticket to a GW. The SIR process is a series of steps to be followed prior to the GW implementing the N/CCO. The SIR will have to be approved and signed by both the Iridium VP of Operations and Planning and SCS Operations Manager (Motorola). See the statements below in order to understand what steps to follow:
 - ☐ The SIR *HAS* been pre-approved? – This means the SNOC personnel have had prior knowledge of this N/CCO, and have already submitted the SIR for approval and received authorization signatures. Proceed to Step 2.7.6 below and continue the normal process.

- ❑ The SIR *HAS NOT* been pre-approved – 1) Create Child Ticket-MO (See Steps 2.7.7-2.7.9 below for instructions on creating the Child Ticket), 2) Transfer Child Ticket-MO to the intended GW and request the GW provide an install date and time, 3) Child Ticket-MO is received back from GW with the install date/time, 4) Assign to Current Assignee - CM POC (found by using the Change Mgt. Assignee Work Group), 5) Log in as CM POC and put Child Ticket-MO into a 'Hold' state. (See the Net Ops lead for instructions and password). CM POC is the holding area for the ticket while awaiting approval signatures from the Iridium and Motorola managers, 6) CM POC personnel will submit the SIR paperwork for approval signatures. 7) SIR has been signed and given to the Net Ops Lead - a copy of the SIR with the proper approval signatures included will be delivered to the Net Ops Lead to proceed with processing, 8) Inform the Iridium Lead of the SIR being received, 9) Continue to Step 6 below. (*NOTE: At this point, you will be skipping Steps 2.7.7, 2.7.8, and 2.7.9 due to the Child Ticket-MO has already been created).
- 2.7.6 FTP the respective files to the indicated GW in the Parent MO - Once approved by the Iridium Lead, you can then FTP the respective files for this N/CCO to the appropriate GW(GTO) (See FTP Operations procedure OP-0201).
- 2.7.7 Open up Parent MO in a Modify Individual window – If you do not already have the Parent MO open in a modify window, do so at this time. (This should already be done from the instructions listed in step 2). You can accomplish this in a number of different ways, but the recommended way is to type in the Parent MO number in the field labeled MT ID, then from the Action Request System line menu select **Query → Modify Individual**.
- 2.7.8 Click on *Create Child MT* – Once you have the Parent MO open in a modify window, click on the Trouble/Maintenance radio button at the bottom of the schema. Under Ticket Association sub- section, click on *Create Child MT* button. This will automatically populate another window in the Submit mode.
- 2.7.9 Create Child MO – In order to successfully create the Child Ticket-MO, you must fill in all the required fields (indicated by bold face typeset) with the same data as the Parent Ticket-MO. Remember, you only paste into the Work Log Description the actual text of the Parent MO (contents of the N/CCO). *NOTE: it is mandatory for all Child Ticket-MO's created that you add this canned message at the beginning in the Work Log Description: *"Please call the SNOC at 703-724-8400 prior to and immediately following the implementation of this N/CCO."*
- 2.7.10 Copy to Submit feature – In order to decrease the time taken in creating multiple Child Ticket-MO's for the same N/CCO, you may utilize the Copy to Submit feature in MTTTS. Once you have created the initial Child Ticket-MO, from the steps listed above, open that same Child Ticket-MO in a Modify Individual window (see instructions listed in *Section 2.5 – Querying the SNOC Database for a MTTTS Ticket*). From the tool line menu at the

top of the Modify Individual window, choose **Actions → Copy to Submit**. At this time a separate window will be displayed in a Submit mode (notice Submit will be listed at the top of the new window being displayed). This new window will be an exact duplicate of the initial Child Ticket-MO that you have already created except for entering data into the Work Log Description and Transferring the ticket to the intended GW. From this new Submit window, the user will need to accomplish the following tasks: 1) Copy and paste the text of the N/CCO into the Work Log Description (under the Trouble/Maintenance section), 2) Under the Transfer section – choose the radio button next to Transfer? Yes, 3) Transfer Type: choose 01-Transfer MT with Ownership, 4) Transfer Destination: choose the appropriate GW that this Child Ticket-MO is intended to be transferred to.

- 2.7.11 Transfer the Child MO – Once you have completed filling in all the data in the required fields, you will now need to transfer this Child MO to the appropriate GW (GTO). You can determine this from the text within the Parent MO. Most of the time the GWs shortname abbreviation is included in the filename of the file(s) that need to be ftp'd. Transferring the Child MO to the GW is the media to inform the GW of action that needs to be taken. From the Submit window, click on the Transfer radio button at the bottom of the Schema. (Transfer of a ticket moves the master copy of the ticket you created from the SNOG server to the server you choose from the menu). Under the Ticket Association section, click the radio button next to Yes under Transfer?, now click the menu list button under Transfer Type: and choose 01-Transfer MT with Ownership, then click the menu list button under Transfer Destination: and from the list displayed choose the appropriate GW that this Child MO is designated to go to. Once you have verified all your data is correct, you can then click the 'Apply' button to create and transfer this Child MO.
- 2.7.12 NOTE: Upon successful creation and transfer of the Child MO, you will receive a confirmation message back from the MTTTS application informing you of the ticket number just created and that it was successfully transferred to the intended GW chosen.
- 2.7.13 Fill out New CCO/MT Log – To keep track and a record of the Parent MO's and Child MO's that have been created, you will need to fill in all the blocks on this log and present this to the Iridium operators.
- 2.7.14 Add a note of the Child MO in the Work Log Description of the Parent MO – Now that the Child MO has been created and transferred to the GW, you will need to add a note in the Work Log Description of the Parent MO, of the Child MO that was created and the GW(GTO) that it was transferred to. While still in the Modify Individual screen of the Parent MO, click on the Trouble/Maintenance radio button at the bottom of the schema, under Trouble/Maintenance Identification section click on the Work Log Description (English): diary button. A Diary Entry window will appear, you need to type in your information on the bottom of the two screens. Once you have accomplished this, click on the 'Apply' button at the bottom of that same Diary Entry window. (NOTE: a note needs to be added to inform other users of the work that has been done. Please include the Child MO number and the GW that the Child MO has been sent to).

- 2.7.15 Put Parent MO on 'Hold' status – Once you have included the note in the Work Log Description, as stated in step 12, click on the Activity radio button at the bottom of the schema. Under the Ticket Activity section, look for the MT State: sub-section and click on the button next to Hold. Next, you will need to change the MT Status: to Hold (050-Hold), you may do so by clicking on the menu list button to the right of the box under MT Status field. Secondly, change the Work Performed: code to Implementing MO (0014-Implementing MO). You can do this by clicking on the menu list button to the right of the box under the Work Performed field.
- 2.7.16 Apply changes/updates – Once step 2.7.15 has been completed, you can now click the 'Apply' button from the main Modify Individual window of the Parent MO, to have your ticket updated.

2.8 UPDATE A TROUBLE TICKET(TT)/MAINTENANCE ORDER (MO)

Updating a TT or MO is merely adding known information to the Work Log Description of a ticket. This information could be actions you have taken to fix a problem, updated status of an ongoing problem, reason for an outage, etc.. Updating the Work Log Description is giving the story of this ticket issue from start to finish. The more and concise data you add, the result is a better understanding of the problem and a greater knowledge is gained. In order to update a TT or MO, the Master copy of the ticket must be on the SNOC MTTS server. This means that when you have the intended ticket to be updated open in a Modify Individual window, click on the Transfer button. Under the System Transfer Information sub-section the *Ownership?* field will have the radio button next to Yes chosen. The *Ownership Server* field will reflect mcf_mmts. Any user on the SNOC server can update the Work Log Description of a ticket on the SNOC server, no matter to whom it is assigned. See steps below:

- 2.8.1 Open the ticket in the Modify Individual mode - Input the ticket number needed into the MT ID: field. Then from the ARS tool line menu, select **Query → Modify Individual**. The ticket number that you have entered should automatically populate in a Modify Individual window. (see instructions listed in *Section 2.5 – Querying the SNOC Database for a MTTS Ticket*).
- 2.8.2 Open the Work Log Description – Once you have the ticket open in a Modify Individual window, click at the bottom of the schema on the Trouble/Maintenance button (this will bring this page to the foreground). Under the Trouble/Maintenance Identification sub-section, click on the Diary entry button next to the Work Log Description (English) field. This will cause the main Work Log Description field to automatically populate.
- 2.8.3 Update the Work Log Description – Once you have the Work Log Description Diary Entry screen displayed, position your mouse and click in the bottom window. You should see a cursor flashing which indicates your text can now be added. The information should be professional, concise, and supportive of the original reason the ticket was opened/created.
- 2.8.4 Apply changes to the Work Log Description – Once you have finished inputting the new data into the Diary Entry window of the Work Log Description, click on the ‘Apply’ button at the lower left corner of the Diary Entry window. This step adds your note into this field, but does not make the note permanent to the ticket record until the next step is accomplished.
- 2.8.5 Apply updated information to the TT/MO – Once you have made all the necessary updates to the Work Log Description, and verified they are correct, you will need to make your updates permanent to the ticket record. To do this click on the ‘Apply’ button at the bottom left corner of the Modify Individual window. The data is saved and the new updates have been added to this record. You will also receive a confirmation message from the MTTS application stating the ticket has been updated successfully.

2.9 ASSIGNING A TROUBLE TICKET (TT)/MAINTENANCE ORDER(MO)

Assigning a ticket is much different from transferring a ticket. The main difference is that the master copy of the ticket being worked remains on the SNOC server when you Assign the ticket. A Transfer actually sends that ticket(Master copy) to another location/server. An example of assigning a ticket would be: the Network Operator position opens a ticket for a feeder link contact failure, he/she is not able to address this issue at their level. He/she will change the Assignee Work Group and Current Assignee to reflect the responsible engineer that will be the one to address this problem. The below steps listed assumes the user is working on a pre-existing ticket. See steps below:

- 2.9.1 Open the desired ticket in the Modify Individual window – (See *Section 2.5 - Querying the SNOC Database for a MTTS Ticket* to open a specific ticket). Input the ticket number needed into the MT ID: field. Then from the ARS tool line menu, select **Query →**

Modify Individual. The ticket number that you have entered should automatically populate in a Modify Individual window. If this is a ticket you are in the process of creating, from *Section 2.6 Open (Create) a Trouble Ticket(TT)* or *2.7 Open (Create) a Maintenance Order(MO) – Child Ticket*, you will have the ticket open in a Submit window. You will be able to follow the remaining steps listed below from the Submit window.

- 2.9.2 **Modify the Work Log Description** – This is a required field and data must be entered. Click on the Trouble/Maintenance button at the bottom of the schema, this will bring this page to the front. Within the Trouble/Maintenance Identification sub-section, click on the Work Log Description(English) Diary Entry button next to this field. If this ticket being worked is received from another entity (GW) first add a note into the Work Log Description – Acknowledging ticket. Then continue inputting all the information you know about this issue. When finished, click the ‘Apply’ button within the Work Log Description Diary Entry window.
- 2.9.3 **Modify the Assignee Work Group** – From the Modify Individual window, click on the Activity button at the bottom of this schema. Within the Ticket Activity sub-section, click on the Assignee Work Group menu button. A popup list or popup menu will appear based on your preferences you have set. From the list/menu displayed, choose the applicable Assignee Work Group. In order to correctly choose the Current Assignee, you must first know what Assignee Work Group that person is assigned to. For whatever Assignee Work Group you choose, the Assignees that are part of this Group will appear.
- 2.9.4 **Apply updated changes made to the TT/MO** – Verify all the data entered is correct, and the correct Assignee Work Group and Current Assignee have been chosen. Click the ‘Apply’ button at the bottom left of the Modify Individual window, for your changes to take affect. (If the individual you have chosen from the menu list is logged into their Notification Tool, they will instantly receive a notification once the Apply button has been depressed).

2.10 TRANSFER A TROUBLE TICKET (TT)/MAINTENANCE ORDER

A Transfer is much different than assigning a ticket. By transferring a ticket you are actually transferring the Master copy of a ticket to another server. Once the ticket has been transferred you are not able to make any changes to that ticket. The ticket is now a read-only copy until the Master copy has been returned. SNOC will only see updates made to the ticket by the GW to which SNOC transferred the ticket to Transfer Type: 01-Transferring MT with Ownership. SNOC will not see updates made to the ticket that was transferred back to a GW, Transfer Type: 03-Return MT with Ownership until the ticket is closed. Assigning a ticket keeps the Master copy of the ticket on your current server, and only assigns the ticket to another individual within the SNOC server. See steps below:

- 2.10.1 **Open the desired ticket in the Modify Individual window** – (See *Section 2.5 - Querying the SNOC Database for a MTTTS Ticket* to open a specific ticket). Input the ticket number

needed into the MT ID: field. Then from the ARS tool line menu, select **Query → Modify Individual**. The ticket number that you have entered will automatically populate in a Modify Individual window. If this is a ticket you are in the process of creating, from *Section 2.6 Open (Create) a Trouble Ticket(TT) or 2.7 Open (Create) a Maintenance Order(MO) – Child Ticket*, you will have the ticket open in a Submit window. You will be able to follow the remaining steps listed below from the Submit window.

- 2.10.2 Modify the Work Log Description – This is a required field and data must be entered. Click on the Trouble/Maintenance button at the bottom of the schema, this will bring this page to the front. Within the Trouble/Maintenance Identification sub-section, click on the Work Log Description(English) Diary Entry button next to this field. If this ticket being worked is received from another entity (GW) first add a note into the Work Log Description – Acknowledging ticket. Then continue inputting all the information you know about this issue. When finished, click the ‘Apply’ button within the Work Log Description Diary Entry window.
- 2.10.3 Modify the Current Assignee – Manually delete the information in this field. Manually input “DSO POC” into the Current Assignee field. MTTS is set up that once a ticket is transferred via the DSO option, the ticket is suppose to default to this Current Assignee (DSO POC). Since this field is required, this step is to ensure there is no problem between the sending server and the receiving server.
- 2.10.4 Update the applicable Transfer fields – Click on the Transfer button at the bottom of this schema. This will bring this page to the foreground Within the Ticket Transfer Information sub-section you have three fields that need to be modified in order for the ticket to be transferred correctly. One, Transfer?, second, Transfer Type:, and lastly, Transfer Destination.
- ☐ Transfer? – You will need to click on the radio button next to Yes
 - ☐ Transfer Type: - This field is very important to understand. Please see APPENDIX B to assist in understanding what Transfer Type is to be used in different situations. (*NOTE: Do Not use Transfer Type: 02-Transfer Independent MT or 04-Transfer Read-Only MT Copy. These two transfer types are not used here at the SNOC.
 - ☐ Transfer Destination: - This will be the actual GW that the Master Copy of the ticket is transferred to. You must choose the correct GW from the menu list for action to be taken. Attention to detail is critical in this step to ensure the correct GW receives the ticket and takes action in a timely manner. (NOTE: If the ticket is being Cleared(Returned) to the Originator the Transfer Type will always be 03-Return MT with Ownership. Once you have chosen this entry, you will not have the option to choose which GW to send the ticket to under the Transfer Destination field). See APPENDIX D for explanations for each entry listed in the Transfer Destination field.

- 2.10.5 Apply updated changes made to the TT/MO – Verify all the data entered is correct, and that you have selected the correct GW within the Transfer Destination field. Click the ‘Apply’ button at the bottom left of the Modify Individual window, for your changes to take affect. You will receive in return from the MTTS application two messages. One, stating the ticket has been transferred to XXX (this should match the intended GW you transferred this ticket to), and secondly, the ticket has been updated successfully.

2.11 HOLD STATUS OF A TROUBLE TICKET(TT)/MAINTENANCE ORDER(MO)

The process of Modifying a ticket to a Hold state is a working functionality of the MTTS application, but the complete concept of the Hold state has not been perfected and is still under development. (The concept of putting a TT/MO on hold is to stop the Mean Time to Fix/Repair clock on a specific ticket and to show that action to be taken on this ticket has been delayed. Some reasons for delay may be: waiting on the customer to provide required information for troubleshooting their problem, a Parent Ticket-MO on hold while GW is implementing the Child Ticket, following a process that requires external approval prior to taking the next plan of action, etc.). In order to change the state of a ticket to Hold, the ticket must be assigned to you(your name in the Current Assignee field). Once you have the information needed to take the next plan of action, you will need to change the MT Status back to its original state prior to being put on Hold. See steps below:

2.11.1 Putting a ticket on ‘Hold’:

- 2.11.1.1 Open the desired ticket in the Modify Individual window – (See *Section 2.5 - Querying the SNOG Database for a MTTS Ticket* to open a specific ticket). Input the ticket number needed into the MT ID: field. Then from the ARS tool line menu, **select Query → Modify Individual**. The ticket number that you have entered will automatically populate in a Modify Individual window. Having the ticket open in a Modify Individual window is the key in order to take action on this ticket.
- 2.11.1.2 Modify the Work Log Description – This is a required field and data must be entered. Click on the Trouble/Maintenance button at the bottom of the schema, this will bring this page to the front. Within the Trouble/Maintenance Identification sub-section, click on the Work Log Description(English) Diary Entry button next to this field. If this ticket being worked is received from another entity (GW) first add a note into the Work Log Description – Acknowledging ticket. Then continue updating the notes of why this ticket is being put on Hold. Be as informative as possible. When finished, click the ‘Apply’ button within the Work Log Description Diary Entry window.
- 2.11.1.3 Modify the MT State field – Click on the Activity button at the bottom of the schema, this will bring this page to the front. Within the Ticket Activity sub-section you will find the MT State field, click on the radio button next to Hold. The radio button is now indented and the button turns yellow.
- 2.11.1.4 Modify the MT Status field – From this same screen as mentioned above in Step 2.11.1.3 and in the Ticket Activity sub-section, click on the menu list button under the

MT Status field. From the list that appears, choose status code 050-Hold, once chosen this code will automatically populate in this field.

- 2.11.1.5 Modify the Work Performed field – NOTE: this field will need to be determined by the user due to the many different scenarios that are possible. The user will need to choose the code that fits the reason this ticket is being put on Hold. (If you are putting the MO-Parent Ticket on Hold, a likely code to use would be 0014-Implementing MO).
- 2.11.1.6 Apply updated changes made to the TT/MO – Verify all the changes you have made are correct, and click the ‘Apply’ button at the bottom left of the Modify Individual window, for your changes to take affect. You will receive in return from the MTTS application a message stating the ticket has been updated successfully.

2.11.2 Taking the ticket off of ‘Hold’ and returning ticket to its original state:

- 2.11.2.1 Open the desired ticket in the Modify Individual window – (See *Section 2.5 - Querying the SNOc Database for a MTTS Ticket* to open a specific ticket). Input the ticket number needed into the MT ID: field. Then from the ARS tool line menu, select **Query → Modify Individual**. The ticket number that you have entered will automatically populate in a Modify Individual window. Having the ticket open in a Modify Individual window is the key in order to take action on this ticket.
- 2.11.2.2 Modify the Work Log Description – This is a required field and data must be entered. Click on the Trouble/Maintenance button at the bottom of the schema, this will bring this page to the front. Within the Trouble/Maintenance Identification sub-section, click on the Work Log Description(English) Diary Entry button next to this field. The difference now will be to update the notes why you are taking the ticket off of Hold and your next plan of action. When finished, click the ‘Apply’ button within the Work Log Description Diary Entry window.
- 2.11.2.3 Modify the MT State field – Click on the Activity button at the bottom of the schema, this will bring this page to the front. Within the Ticket Activity sub-section you will find the MT State field, click on the radio button next to Open. The radio button is now indented and the button turns yellow. The radio button next to Hold will no longer be indented or show color of yellow.
- 2.11.2.4 Modify the MT Status field – From this same screen as mentioned above in Step 2.11.2.3 and in the Ticket Activity sub-section, click on the menu list button under the MT Status field. From the list that appears, choose status code 000-Creating Ticket. (NOTE: If the action required for this ticket has been taken and you are returning the ticket, you will need to Clear the ticket back to the originator. See *Section 2.12 – Clear a Trouble Ticket(TT)/Maintenance Order(MO) to the Originator*, steps 2.12.2.3 – 2.12.2.6).

2.11.2.5 Modify the Work Performed field – NOTE: this field will need to be determined by the user due to the many different scenarios that are possible and the code used should reflect the action that was taken. (NOTE: If the action required for this ticket has been taken and you are returning the ticket, you will need to Clear the ticket back to the originator. See *Section 2.12 – Clear a Trouble Ticket(TT)/Maintenance Order(MO) to the Originator*, steps 2.12.3 – 2.12.8).

2.11.2.6 Apply updated changes made to the TT/MO – Verify all the changes you have made are correct, and click the ‘Apply’ button at the bottom left of the Modify Individual window, for your changes to take affect. You will receive in return from the MTTS application a message stating the ticket has been updated successfully.

2.12 CLEAR A TROUBLE TICKET(TT)/MAINTENANCE ORDER(MO) TO THE ORIGINATOR

Clearing a ticket is very similar to transferring a ticket, in that you are transferring the Master Copy of the ticket back to the Originator, updating the status of the ticket to reflect the status of being cleared, and informing of the action(s) taken for resolution. Clearing the ticket informs the issue has been resolved to the best of your knowledge. In MTTS, the act of returning a TT/MO to its originator is termed ‘CLEAR’. (NOTE: As mentioned in previous sections, there are several required fields that must be modified, in order to complete this action. The required fields are: *Work Log Description, MT Status, Work Performed, Current Assignee, Cause, and Resolution*). Although Cause and Resolution fields are not bold face typeset, whenever you change the MT Status to 200-Cleared, these fields then become required through an Active Link that has been pre-defined. Other fields that need to be updated for accurate report recording are: Final System/Service and Final Affected Element Id. These fields are not required (bold face typeset), but need to be updated to reflect the Final element that was worked on and what action was taken to resolve this issue).

Examples of this would be:

- ❑ A Trouble Ticket (TT) has been received requesting information or action to be taken by the SNOC personnel (GW requests SV Element sets for a specific date be ftp’d to them). The Real-time Network Operations take the appropriate action and fulfill the request, update the ticket accordingly and clear the ticket back to the GW.
- ❑ A Maintenance Order (MO-Parent ticket) has been received from GTAC or NIP requesting a change to a specific GW office data, software, or hardware. A Child ticket is created and sent to the GW, GW takes action and clears the ticket back to SNOC. SNOC closes the Child ticket opened for this action, updates the Work Log Description in the Parent ticket, changes the appropriate codes and clears the MO-Parent ticket back to the Originator.

- 2.12.1 Open the TT/MO in a Modify Individual window - (See *Section 2.5 - Querying the SNOC Database for a MTTS Ticket* to open a specific ticket). Input the ticket number needed into the MT ID: field. Then from the ARS tool line menu, select **Query → Modify Individual**. The ticket number that you have entered will automatically populate in a Modify Individual window.
- 2.12.2 Modify the Work Log Description – This is a required field and data must be entered. Click on the Trouble/Maintenance button at the bottom of the schema, this will bring this page to the front. Within the Trouble/Maintenance Identification sub-section, click on the Work Log Description(English) Diary Entry button next to this field. If this ticket being worked is received from another entity (GW) first add a note into the Work Log Description – Acknowledging ticket. Then continue inputting all the information on the actions that have been taken. When finished, click the ‘Apply’ button within the Work Log Description Diary Entry window.
- 2.12.3 Modify the MT Status Field – Once you have included the note in the Work Log Description, as stated in step 2.12.2, click on the Activity button at the bottom of the schema. Within the Ticket Activity sub-section, click on the menu list button next to the MT Status field. From the list that now appears, change the MT Status to Cleared (200-Cleared).
- 2.12.4 Modify the Work Performed field – While still in the Ticket Activity sub-section (as mentioned above in Step 2), click on the menu list button next to the Work Performed field. From the list that now appears, change the Work Performed to 0005-Clearing Problem. By changing this field, this action makes the Cause and Resolution fields required (see step 2.12.5).
- 2.12.5 Modify the Cause and Resolution fields – Within the MTTS application, there is an active link. This Active Link has been designed that once you change the Work Performed field to 0005-Clearing Problem, the Cause and Resolution fields then become required fields. In updating these two fields, you are accurately reporting what the actual cause of the outage was and what the resolution was to fix this issue. It is the responsibility of the user, that is clearing this ticket, to use the exact codes of the action taken to fix this problem. Proper use of codes will assist in analyzing data and problem areas, and know where to focus man hours toward eliminating problems. If for some reason, the codes you need to use are not listed, notify the MTTS Administrator as soon as possible via e-mail or phone call.
- 2.12.6 Modify the Current Assignee – Manually delete the information in this field. Manually input “DSO POC” into the Current Assignee field. MTTS is set up that once a ticket is transferred via the DSO option, the ticket is suppose to default to this Current Assignee (DSO POC). Since this field is required, this step is to ensure there is no problem between the sending server and the receiving server.

2.12.7 Update the applicable Transfer fields – Click on the Transfer button at the bottom of this schema. This will bring this page to the foreground. Within the Ticket Transfer Information sub-section the user will need to modify two fields (see below):

- ☐ Transfer? – You will need to click on the radio button next to Yes
- ☐ Transfer Type: - Click on the menu list button under the Transfer Type field and choose 03-Return MT with Ownership. (This will automatically route the ticket back from where it came). Please see APPENDIX B to assist in understanding what Transfer Type is to be used in different situations.

2.12.8 Apply updated changes and Clear the TT/MO back from wherever the ticket was delivered from. Verify all the data entered is correct, and that the correct Transfer Type code has been chosen. Click the 'Apply' button at the bottom left of the Modify Individual window, for your changes to take affect and the ticket to be transferred back to where it came from. You will receive, in return from the MTTS application, two messages upon a successful clearing of the ticket. One, stating the ticket has been transferred to XXX. Secondly, a message will state the ticket has been updated successfully.

2.13 CLOSE A TROUBLE TICKET(TT)/MAINTENANCE ORDER(MO)

Closing a TT or MO is an action that is taken to inform the original reason for the creation of this TT/MO has been resolved. Only the Originator or Originator's Manager may take action to Close a TT/MO(within the SNOC environment the Administrator has set up entities that have manager capabilities in the absence of the Originator). The Originator of the ticket must also ensure this ticket has been Cleared prior to him or her Closing the ticket. If this has not been done, See *Section 2.12 - Clear a Trouble Ticket(TT)/Maintenance Order(MO) to the Originator*. See steps below:

- 2.13.1 Open the TT/MO in a Modify Individual window - (See *Section 2.5 - Querying the SNOOC Database for a MTTTS Ticket* to open a specific ticket). Input the ticket number needed into the MT ID: field. Then from the ARS tool line menu, select **Query → Modify Individual**. The ticket number that you have entered will automatically populate in a Modify Individual window. This step assumes you have already received notification of the ticket being assigned to you from the Notification Tool.
- 2.13.2 Modify the Work Log Description – This is a required field and data must be entered. Click on the Trouble/Maintenance button at the bottom of the schema, this will bring this page to the foreground. Within the Trouble/Maintenance Identification sub-section, click on the Work Log Description(English) Diary Entry button next to this field. The user must add a descriptive explanation of the work that was performed in order to resolve this issue and that he/she is Closing the ticket. (See Appendix C to ensure all the proper information has been entered into the Work Log Description prior to closing the ticket). When finished, click the ‘Apply’ button within the Work Log Description Diary Entry window.
- 2.13.3 Modify the MT State Field – Once you have included the note in the Work Log Description, as stated in step 2.13.2, click on the Activity button at the bottom of the schema. Within the Ticket Activity sub-section and under the MT State field, click on the radio button next to Closed. The radio button will automatically become indented and turn yellow to indicate the selection you have chosen.
- 2.13.4 Modify the MT Status Field – Once you have verified this ticket has been successfully Cleared, this field will remain the same in the action being taken to Close the ticket. If the ticket has not been cleared follow the instructions listed in *Section 2.11 - Clear a Trouble Ticket(TT)/Maintenance Order(MO) to the Originator*.
- 2.13.5 Modify the Work Performed Field – While still in the Ticket Activity sub-section (as mentioned above in Step 3), click on the menu list button next to the Work Performed field. From the list that now appears, change the Work Performed code to 0006-Closing Ticket. (As stated before, prior to closing the ticket the ticket should have been properly cleared. If this has not been done, follow the instructions listed in *Section 2.11 - Clear a Trouble Ticket(TT)/Maintenance Order(MO) to the Originator*).
- 2.13.6 Apply the updated changes to the TT/MO – Verify all the data entered and changes made are correct. Click the ‘Apply’ button at the bottom left of the Modify Individual window, for your changes to take affect and the ticket to be Closed. You will receive, in return from the MTTTS application, a message stating the ticket has been successfully closed and the Originator has been updated of the ticket closure. You will receive this message upon successful Closure of the ticket.

2.14 RESETTING TICKETS IN A HUNG STATE

This section will attempt to explain what constitutes a ticket in a hung state and the process for resetting a hung ticket.

- ❑ **What is a hung ticket?** - A hung ticket is when an operator has attempted to Transfer a ticket (TT or MO) to another entity outside of the SNOC server and for many possible reasons this operation did not process successfully. The key fields to verify are the Ownership? and the Ownership Server. When a Transfer of a ticket has successfully completed the Ownership? will have the radio button next to 'No' highlighted, and the Ownership Server will show the GW that the ticket was Transferred to (i.e.: omcg.gw01.iwcs.net). If the operator has attempted to Transfer the ticket and later has gone in to check the status of this ticket and the key fields to verify do not show a successful Transfer, the ticket is most likely hung. (*NOTE: The best way to verify the location of the master copy of the ticket is to input data into the Work Log Description and click on the 'Apply' button at the bottom of the schema. If you receive a message in return stating 'Ticket has been modified successfully' the master copy of the ticket is still on the SNOC server and the Transfer did not work. If you receive a message stating 'Cannot update this entry-this distributed entry is not the master copy' then the master copy of the ticket is residing on the GW server). You may see the Ownership? field showing 'Yes' but the Ownership Server field showing omcg.gw01.net.iwcs (or vice versa) – this is also a likely condition of a hung ticket. Again, the best way to verify is to update the Work Log Description and click on 'Apply' to see what message you receive in return.
- ❑ **How to reset a Hung Ticket** – After verifying that the current ticket you are working on is in fact in a Hung state (see above rules), follow these steps: 1) Open a new terminal window – from the Unix main work area, bring up a new Terminal window, 2) Type `xhost +mcf_mmts` (there is a space between xhost and +mcf_mmts) and press Return, 3) From the command prompt type `>telnet mmts`, press Return (this will provide you a login for access into the mcf_mmts server), 4) At the login prompt type: `cfso` (lowercase), 5) At the Password prompt, type: `cfso` (lowercase). You will receive a `cfso` command prompt in return and you are logged into the server with limited Administrator privileges. 6) From the `cfso` command prompt, type: `reset_ticket XXX00000XXXX` (NOTE: the ticket number must be exactly the way it is displayed in MTTTS or this procedure will not work). 7) The system will then display(echo) back the command you have entered and the information *Action Request System Distributed Server Admin version 3.0* and the copyright information and then the `cfso` command prompt. If there is no error message displayed right above the `cfso` command prompt that is returned, then the ticket has been successfully reset. If you do receive an error message – contact the MTTTS System Administrator. 8) Once the ticket has been reset, you may now take action on the ticket and/or attempt the Transfer of the ticket to a specific GW again.

APPENDIX A

List of MTTs Codes Used to Create TT/MO's

SV ANOMALY:

<u>Reason</u>	<u>System/Service</u>	<u>Subsystem/Feature</u>	<u>Trbl/Maint Type:</u>
No RF from SV	0000-SV	0000	0021-0024
Failed To sched	0000-SV	0000	0055
SV deleted to support another SV	0000-SV	0002	0009
SVARC switch (planned/unplanned)	0000-SV	0000	0014/0015
Coldstarts/Quickstarts	0000-SV	0000	0014 – 0020
Main Mission Antenna	0000-SV	0000	0010 – 0013
Ant Comp	0000-SV	0000	0017 – 0020

TTAC:

<u>Reason</u>	<u>System/Service</u>	<u>Subsystem/Feature</u>	<u>Trbl/Maint Type:</u>
ETS Holdoff	0000-SV	0000	0007
High Power Amplifier	2000	0002 – 0005 (ET)	0012
Hardware Failure (applicable)	2000	0002 – 0005 (ET)	0003 – 0015
Software Failure	2000	0009	0003

MTC:

<u>Reason</u>	<u>System/Service</u>	<u>Subsystem/Feature</u>	<u>Trbl/Maint Type:</u>
ISRP stops/hung	1000	0024	0001
MTC Switchover failure	1000	0011	0001
LAN failure (fddi/Internet (tnpp))	1000	0026	0001
Com Services	1000	0014	0001
Corrupt Tables	1000	0027	0002

MOC:

<u>Reason</u>	<u>System/Service</u>	<u>Subsystem/Feature</u>	<u>Trbl/Maint Type:</u>
Alarms/Unable to reset	1000	0013	0001

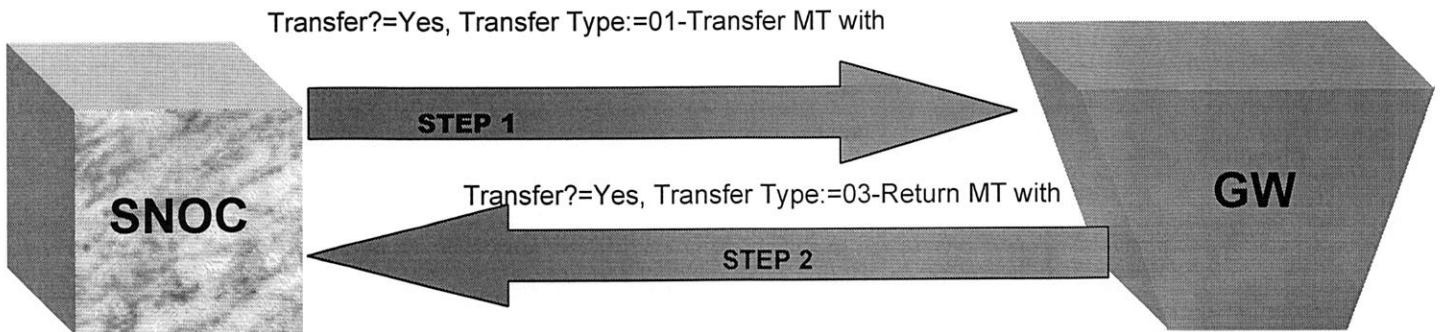
ODN:

<u>Reason</u>	<u>System/Service</u>	<u>Subsystem/Feature</u>	<u>Trbl/Maint Type:</u>
ODN Outage	1000	0000	0004

APPENDIX B

Illustration of a Transfer vs. Assigning a TT/MO

Flow of a Trouble Ticket Transferred to a

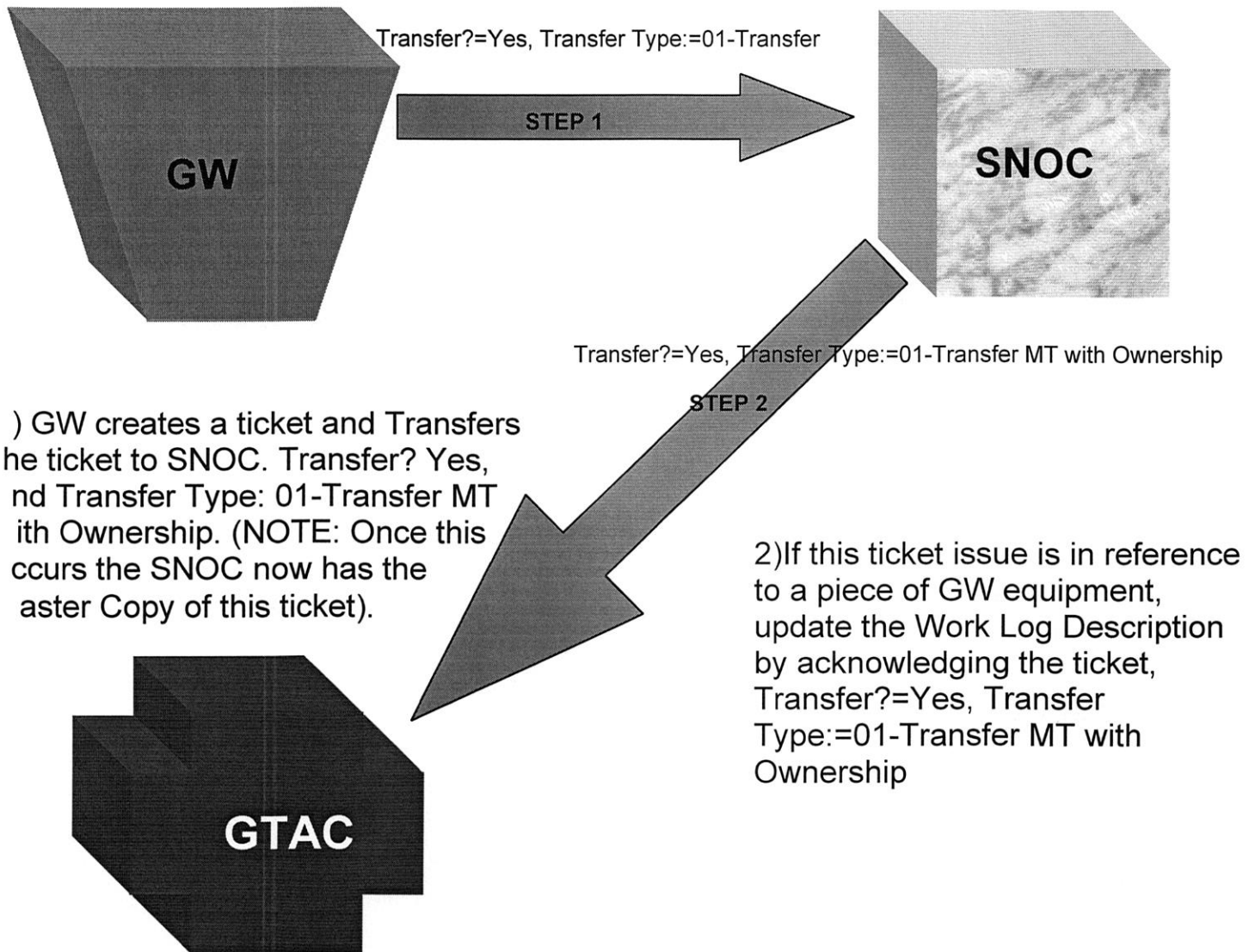


1) Create ticket and Transfer ticket to GW. Transfer? Yes, and Transfer Type: 01-Transfer MT with Ownership. (NOTE: this transfers the Master Copy of the ticket to the GW Server(OMCG))

2) GW takes action and updates the ticket. GW should clear and transfer the ticket back: Transfer?Yes, Transfer Type: 03-Return MT with Ownership

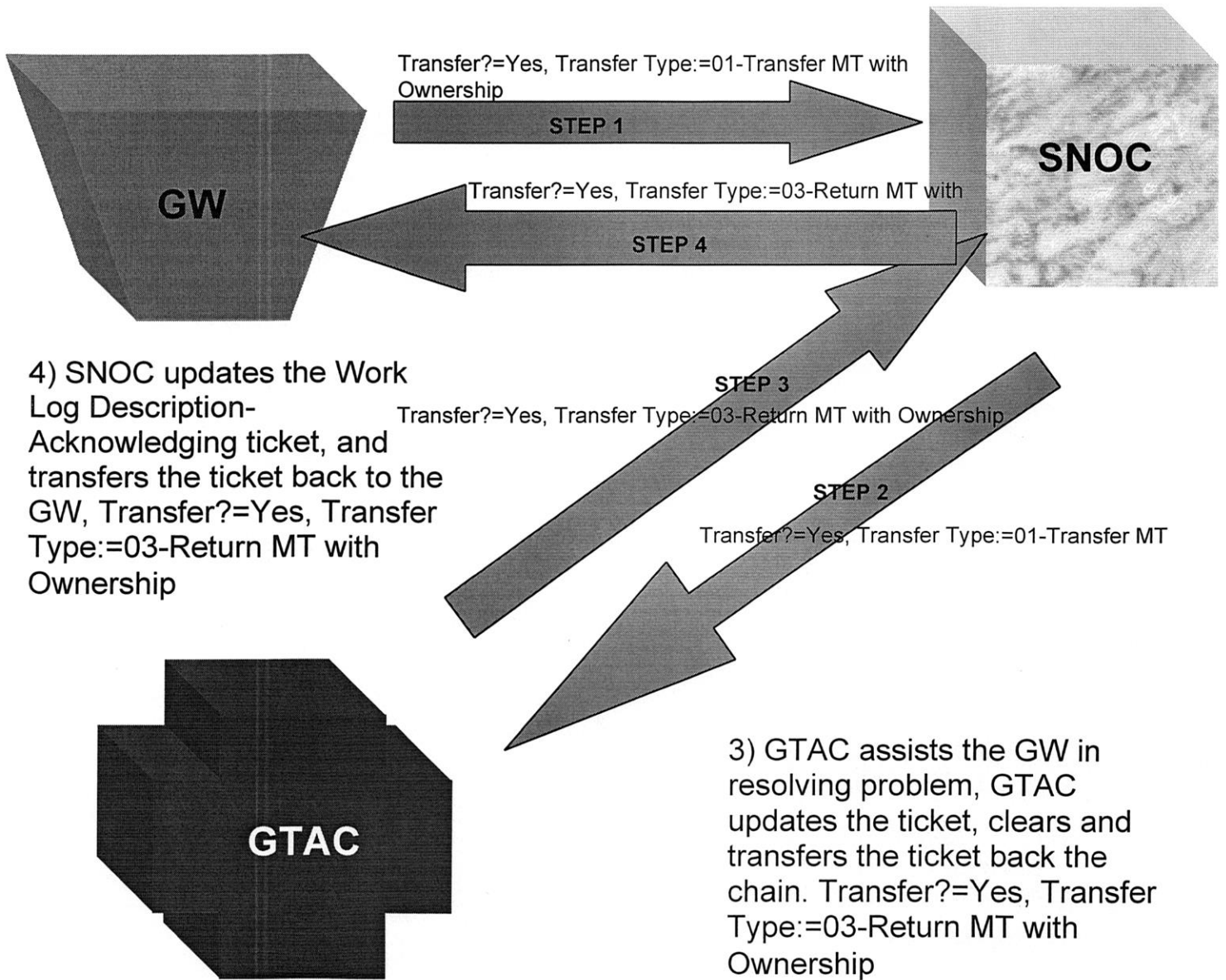
*NOTE: This is the same flow of operation when a ticket is sent from the GW to SNOC, then back to the GW.

APPENDIX B (CONT.)



APPENDIX B (CONT.)

Flow of a GW created Trouble ticket (GW, SNOC, GTAC)



APPENDIX B (CONT.)

NOTE: When you are transferring a ticket to another entity and you expect to receive this ticket back with updates, you will always use Transfer Type: 01-Transfer Ticket with Ownership.

If you are returning a ticket that has come to the SNOC server and SNOC did not originate the ticket you will always use Transfer Type: 03-Return MT with Ownership

NOTE: When you Transfer a ticket to another entity, the Master copy of that ticket is also transferred. Only a copy of the ticket remains on the SNOC server (until it is returned, then DSO POC is notified).

APPENDIX C

Actions to be taken on a ticket in step format

SECTION 2.5 – QUERYING THE SNOC DATABASE FOR A MTTT TICKET

STEP#	Action to be taken
1	Ensure you have completed Section 2.2 and have access into the MTTT application
2	*Note: This operation can be done in one of two ways: First – Enter syntax specific data into the Query bar at the bottom of the Main ARS screen ('MT ID:' LIKE "IMCF0000XXXXX" or... Second – Input data into a specific field. Put IMCF0000XXXXX into the open field under MT ID:
3	Go to the ARS tool line menu, select Query → List (for a quick brief description of the ticket in a List mode) or Modify Individual in order to make update/changes to the ticket. *Note: Having the ticket open in a Modify Individual window is the key to updating and applying changes to any ticket on the SNOC server.

SECTION 2.6 – OPEN (CREATE) A TROUBLE TICKET (TT)

STEP#	Action to be taken
1	Ensure you have completed Section 2.2 and have access into the MTTT application
2	From the ARS window tool line menu, select File → Open Submit (A new window will be displayed in the Submit mode).
3	Input data into the required fields of the Submit window. (Required fields are: <i>Originator, System/Service, Subsystem/Feature, Trouble/Maintenance Type, Work Log Description(English), Initial Affected Element ID, Priority, MT State, MT Status, Work Performed, and Current Assignee</i>). See APPENDIX A for correct codes to be used for each scenario. *Note: This data entered into the above listed fields must reflect the reason for opening the ticket
4	-If ticket is destined for another entity outside of the SNOC server – Transfer the ticket (follow instructions listed in Section 2.10). -If ticket is destined for an engineer within the SNOC server – Assign the ticket (follow instructions listed in Section 2.9).
5	Click on the 'Apply' button at the bottom of the Submit window to initiate the creation of a ticket. *Note: The MTTT application will then return a message stating the ticket number that has been created.

APPENDIX C (CONT.)

SECTION 2.7 – OPEN (CREATE) A MAINTENANCE ORDER(MO) – CHILD TICKET

STEP#	Action to be taken
1	Ensure you have completed Section 2.2 and have access into the MTTTS application
2	Acknowledge incoming Parent ticket-Open the Parent Ticket-MO in a Modify Individual window. Add a note into the Work Log Description – “Acknowledging ticket.”
3	Verify Parent ticket-MO is in the correct format and proper approving signatures are included. (If user is not sure of this format consult with the Motorola or Iridium Lead Tech).
4	Inform Iridium Lead Tech or designated representative of the MO being received.
5	Obtain approval from Iridium Lead Tech or designated representative to FTP the corresponding files for this N/CCO to the appropriate GW.
6	FTP the respective files to the appropriate GWs listed in the text of the Parent Ticket-MO received from GTAC or NIP. (See FTP Operations procedure OP-0201).
7	Open Parent Ticket-MO in a Modify Individual window
8	Click on Create Child MT button. This button is located under the Trouble/Maintenance button from the main schema window, and within the Ticket Association sub-section. *Note: This will automatically populate a new window in the Submit mode.
9	Input correct data into all the required fields: <i>Originator, System/Service, Subsystem/Feature, Trouble/Maintenance Type, Work Log Description(English), Initial Affected Element ID, Priority, MT State, MT Status, Work Performed, and Current Assignee</i>).
10	Transfer Child Ticket-MO to the appropriate GW that the related files have been FTP'd to. (For understanding of a Transfer see Section 2.10 or APPENDIX B). *Note: Upon successful creation and transfer of the Child Ticket-MO you will receive a message stating the ticket number that was created and the ticket has been sent to the GW designated.
11	Fill out a new CCO/MT Log and present this to the Iridium Lead Tech or designated representative. This is to show the work accomplished and as a checking system of sending the files and ticket to the correct GW. The Iridium operators will then log this information into their database for implementation tracking.
12	Put Parent Ticket-MO on 'Hold' status and update the Work Log Description with the Child Ticket number(s) you have created and the GWs each ticket was sent to.
13	Click on the 'Apply' button at the bottom of the Modify Individual window of the Parent Ticket-MO. This will make your updates permanent to the Parent Ticket-MO

APPENDIX C (CONT.)

SECTION 2.8 – UPDATE A TROUBLE TICKET(TT)/MAINTENANCE ORDER(MO)

STEP#	Action to be taken
1	Ensure you have completed Section 2.2 and have access into the MTTTS application
2	Open the intended TT or MO in a Modify Individual window in order to make updates to the ticket in the database. (See Section 2.5 for instructions on doing this).
3	Update the Work Log Description – click on the Trouble/Maintenance button at the bottom of the schema, then click on the Diary Entry button next to the Work Log Description field within the Trouble/Maintenance Identification sub-section.
4	Apply your updates to the Work Log Description – click on the ‘Apply’ button at the lower left corner of the Diary Entry Window within the Work Log Description
5	Click on the ‘Apply’ button at the bottom of the Modify Individual window in order for your updates to take affect to the ticket record in the database.

SECTION 2.9 – ASSIGNING A TROUBLE TICKET(TT)/MAINTENANCE ORDER(MO)

STEP#	Action to be taken
1	Ensure you have completed Section 2.2 and have access into the MTTTS application
2	Open the intended TT or MO in a Modify Individual window in order to make updates to the ticket in the database. (See Section 2.5 for instructions on doing this).
3	Modify the Work Log Description – Click on the Trouble/Maintenance button at the bottom of the schema, then click on the Diary Entry button next to the Work Log Description field within the Trouble/Maintenance Identification sub-section. Input into the notes to whom you are assigning this ticket and why you are assigning the ticket to them.
4	Modify the Assignee Work Group – Choose from the list that appears the correct Assignee Work Group, that the person you want to assign the ticket to, belongs to
5	Modify the Current Assignee – This will be the person to whom you want to take action on this ticket. The list that appears will be determined from the Assignee Work Group that you have chosen in Step 4.
6	Click on the ‘Apply’ button at the bottom of the Modify Individual window in order for your updates to take affect to the ticket record in the database. *Note: The instant that you click the ‘Apply’ button the individual that you have assigned the ticket to will receive a notification of assignment on their Notification Tool.

APPENDIX C (CONT.)

SECTION 2.10 – TRANSFER A TROUBLE TICKET(TT)/MAINTENANCE ORDER(MO)

STEP#	Action to be taken
1	Ensure you have completed Section 2.2 and have access into the MTTTS application
2	Open the intended TT or MO in a Modify Individual window in order to make updates to the ticket in the database. (See Section 2.5 for instructions on doing this).
3	Modify the Work Log Description – Click on the Trouble/Maintenance button at the bottom of the schema, then click on the Diary Entry button next to the Work Log Description field within the Trouble/Maintenance Identification sub-section. Input into the notes your reason for Transferring the ticket and to which GW you have intended the ticket to be sent to.
4	Modify the Current Assignee field – manually input into this field DSO POC. Since the default for routing between server locations is DSO POC, this extra step is to ensure system integrity.
5	Modify the applicable Transfer fields: <i>Transfer?</i> , <i>Transfer Type:</i> , and <i>Transfer Destination:</i> . *Note: If the ticket is being Cleared(Returned) to the Originator the Transfer Type will always be 03-Return MT with Ownership. Once you have chosen this entry, you will not have the option to choose which GW to send the ticket to under the Transfer Destination field.
6	Click on the ‘Apply’ button at the bottom of the Modify Individual window in order for your updates to take affect and the ticket to be Transferred to the intended GW. *Note: The user will then see a message stating the ticket has been updated successfully and the ticket has been transferred to GW XXX. (The GW listed in this notification message should be the intended GW that you have chosen. If this is not correct the user has either chosen the wrong GW in error, or their may be a problem within the MTTTS application and the mapping files, although the later does not change unless there has been a recent upgrade to the system).

SECTION 2.11 – HOLD STATUS OF A TROUBLE TICKET(TT)/MAINTENANCE ORDER(MO)

STEP#	Action to be taken
*Putting a TT or MO into a ‘Hold’ state:	
1	Ensure you have completed Section 2.2 and have access into the MTTTS application
2	Open the intended TT or MO in a Modify Individual window in order to make updates to the ticket in the database. (See Section 2.5 for instructions on doing this).
3	Modify the Work Log Description – Click on the Trouble/Maintenance button at the bottom of the schema, then click on the Diary Entry button next to the Work Log Description field within the Trouble/Maintenance Identification sub-section. Input into the notes your reason for putting this ticket into a ‘Hold’ state.
4	Modify the MT State field – Click on the radio button next to Hold

APPENDIX C (CONT.)

5	Modify the MT Status field – Click on the menu list button and from the list provided choose the code 050-Hold
6	Modify the Work Performed field – This field should reflect the reason for putting on Hold and is determined by the user. (*Note: An example would be putting a Parent Ticket-MO on hold while the GW is implementing the N/CCO. Click on the menu list button and from the list provided choose the code 0014-Implementing MO).
7	Click on the 'Apply' button at the bottom of the Modify Individual window in order for your updates to take affect and the ticket to go into a 'Hold' state.
*Taking a TT or MO out of a 'Hold' state:	
1	Ensure you have completed Section 2.2 and have access into the MTTS application
2	Open the intended TT or MO in a Modify Individual window in order to make updates to the ticket in the database. (See Section 2.5 for instructions on doing this).
3	Modify the Work Log Description – Click on the Trouble/Maintenance button at the bottom of the schema, then click on the Diary Entry button next to the Work Log Description field within the Trouble/Maintenance Identification sub-section. Input into the notes your reason for taking this ticket out of 'Hold' status.
4	Modify the MT State field – Click on the radio button next to Open
5	Modify the MT Status field – Click on the menu list button and from the list provided choose the code 000-Creating Ticket
6	Modify the Work Performed field – This field should reflect the reason for taking the ticket off of Hold status or what action was performed while on Hold status. *Note: If the user has resolved the problem or has answered the questions asked you will want to Clear the ticket back to its Originator at this point. See Section 2.12 for instructions on doing this.
7	Click on the 'Apply' button at the bottom of the Modify Individual window in order for your updates to take affect and the ticket to go back to its original state.

SECTION 2.12 – CLEAR A TROUBLE TICKET(TT)/MAINTENANCE ORDER(MO) TO THE ORIGINATOR

STEP#	Action to be taken
1	Ensure you have completed Section 2.2 and have access into the MTTS application
2	Open the intended TT or MO in a Modify Individual window in order to make updates to the ticket in the database. (See Section 2.5 for instructions on doing this).
3	Modify the Work Log Description – Click on the Trouble/Maintenance button at the bottom of the schema, then click on the Diary Entry button next to the Work Log Description field within the Trouble/Maintenance Identification sub-section. Input into the notes all the actions you have taken on this ticket, to include the fix action. Then click the 'Apply' button within the Diary Entry window.
4	Modify the MT Status field – Click on the menu list button and from the list provided choose the code 200-Cleared
5	Modify the Work Performed field – Click on the menu list button and from the list provided choose the code 0005-Clearing Problem

APPENDIX C (CONT.)

6	Modify the Cause field – Once the Work Performed field has been changed to 0005-Clearing Problem this makes the Cause field a required entry field. The code used should correctly reflect the action taken for resolution of this problem.
7	Modify Resolution field - Once the Work Performed field has been changed to 0005-Clearing Problem this makes the Resolution field a required entry field. The code used should correctly reflect the action taken for resolution of this problem. *Note: The Final System/Service and Final Affected Element ID fields should also be correctly updated for analysis reporting.
8	Modify the Current Assignee field – manually input into this field DSO POC. Since the default for routing between server locations is DSO POC, this extra step is to ensure system integrity.
9	Update the applicable Transfer fields – Choose the radio button next to Yes, under the Transfer? field. Click on the menu list button under the Transfer Type field and choose 03-Transfer MT with Ownership.
10	Click on the ‘Apply’ button at the bottom of the Modify Individual window in order for your updates to take affect and the ticket to be Cleared (Returned) to the tickets Originator. *Note: The user will then see a message stating the ticket has been updated successfully and the Originator “XXX” has been notified of the ticket being cleared.

APPENDIX D

Transfer Destination Field Explanation Chart

Destination	Explanation
002 – MTS-PRD	This is the Transfer route to send tickets to the Iridium Internet Billing System (IIBS) – this is a Business office (GBO)
003 – MCF_MTTs	This is the route code for SNOc server. This should not be used by SNOc personnel.
004 – BCFMT	This is the Transfer route to send tickets to the Back-up Control Facility. The BCF does not currently have an IP Address or route mapping – this is a Network office (GTO)
005 – GTACMT	This is the Transfer route to send tickets to the Gateway Technical Assistance Center(GTAC). This office is where tickets would be routed for answers/assistance to technical questions on specific GW pieces of equipment – this is a Network office (GTO)
006 – BSCMT	This is the Transfer route for the Business Service Center(BSC) which has merged with TSC and is now called the Iridium Technical Center(ITC). This group acts as a Clearinghouse for all tickets that originate from the GW's and this office determines where to route each specific ticket for assistance or, if possible, resolve the issue in-house – this is a Business office (GBO)
007 – GCCMT	This is the Transfer route for the Global Customer Care Center (GCC). This office is a call in point for all Iridium customer support issues – this is a Business office (GBO)
024 – GTO US	This is the Transfer route for the Gateway Technical Office (GTO) located in Arizona. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)
026 – GTO Japan	This is the Transfer route for the Gateway Technical Office (GTO) located in Japan. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)
027 – GTO Thailand	This is the Transfer route for the Gateway Technical Office (GTO) located in Thailand. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)
028 – GTO Russia	This is the Transfer route for the Gateway Technical Office (GTO) located in Russia. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)
029 – GTO Korea	This is the Transfer route for the Gateway Technical Office (GTO) located in Korea. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)

APPENDIX D (CONT.)

030 – GTO Italy	This is the Transfer route for the Gateway Technical Office (GTO) located in Italy. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)
031 – GTO India	This is the Transfer route for the Gateway Technical Office (GTO) located in India. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)
032 – GTO Taiwan	This is the Transfer route for the Gateway Technical Office (GTO) located in Taiwan. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)
033 – GTO Middle East	This is the Transfer route for the Gateway Technical Office (GTO) located in Middle East. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)
034 – GTO China	This is the Transfer route for the Gateway Technical Office (GTO) located in China. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)
035 – GTO Hawaii	This is the Transfer route for the Gateway Technical Office (GTO) located in Hawaii. This office is where all the equipment that interfaces with the Iridium constellation is located, this office is in contrast to the Gateway Business Office (GBO)
037 – NPS-PRD	This is the Transfer route for the office of Network Implementation & Planning (NIP). This is the office that NCCO's originate from – this is a Business office (GBO)
038 – AST-MT	This is the Transfer route for the Application Support Team of MTTS. This routing is currently <i>not</i> being used until further notice.
039 – GTACGW	This is the Transfer route for the GTAC Gateway (GTACGW) test lab. This is a test lab environment where GTAC implements all new software and hardware upgrades prior to releasing them to the Gateways (GTO's) – this is a Network office (GTO)

REVISION HISTORY

12/30/99	B. Faircloth	First draft
[M/d/yy	I. Lastname	Brief description of the change]